The Vibrational Raman Spectra

If a molecule has <u>no symmetry</u>: **all the vibrational modes are Raman active.**

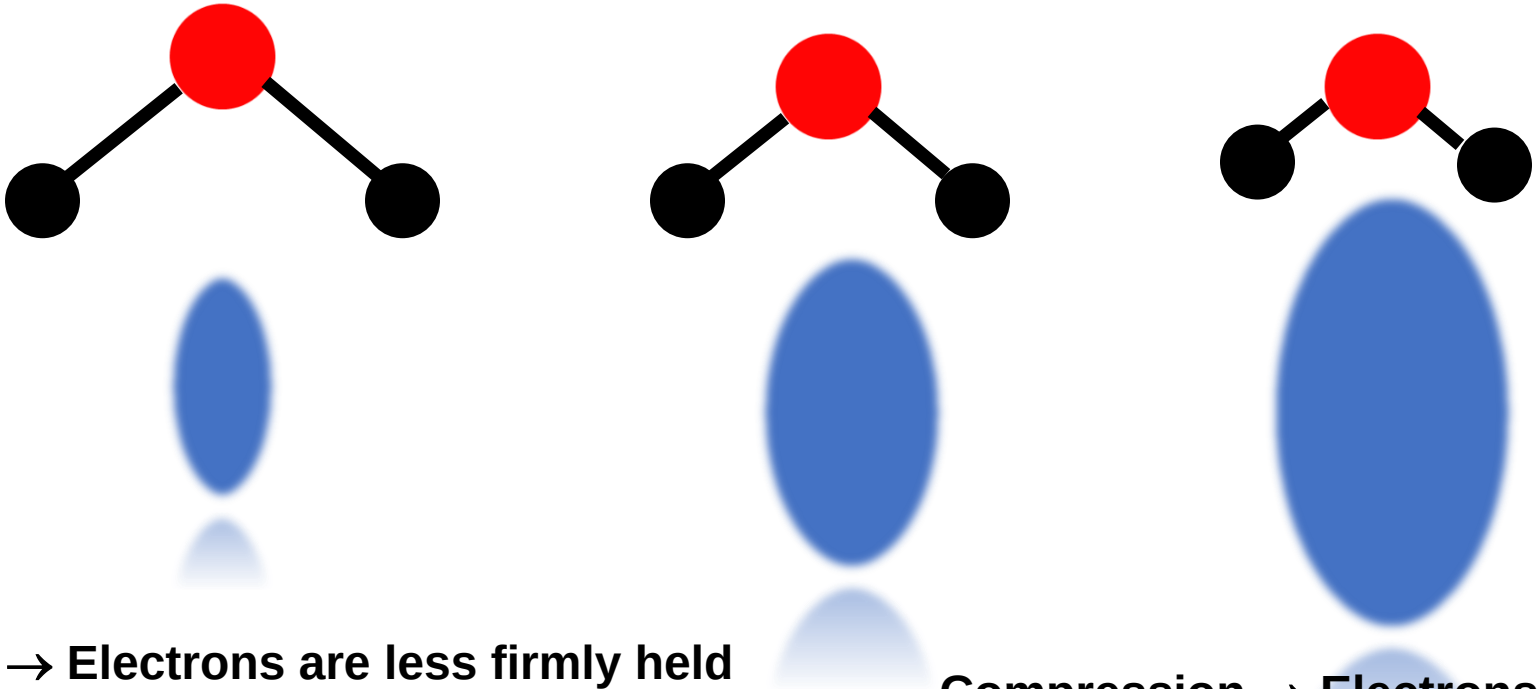
If a molecule has <u>considerable symmetry</u>: **all the vibrational modes may not be Raman active.**

↓

Depends upon whether or not the polarizability changes during vibration.

The Vibrational Raman Spectra

Symmetric stretching mode in water



Stretching → Electrons are less firmly held to the nuclei → Bond becomes more polarizable.

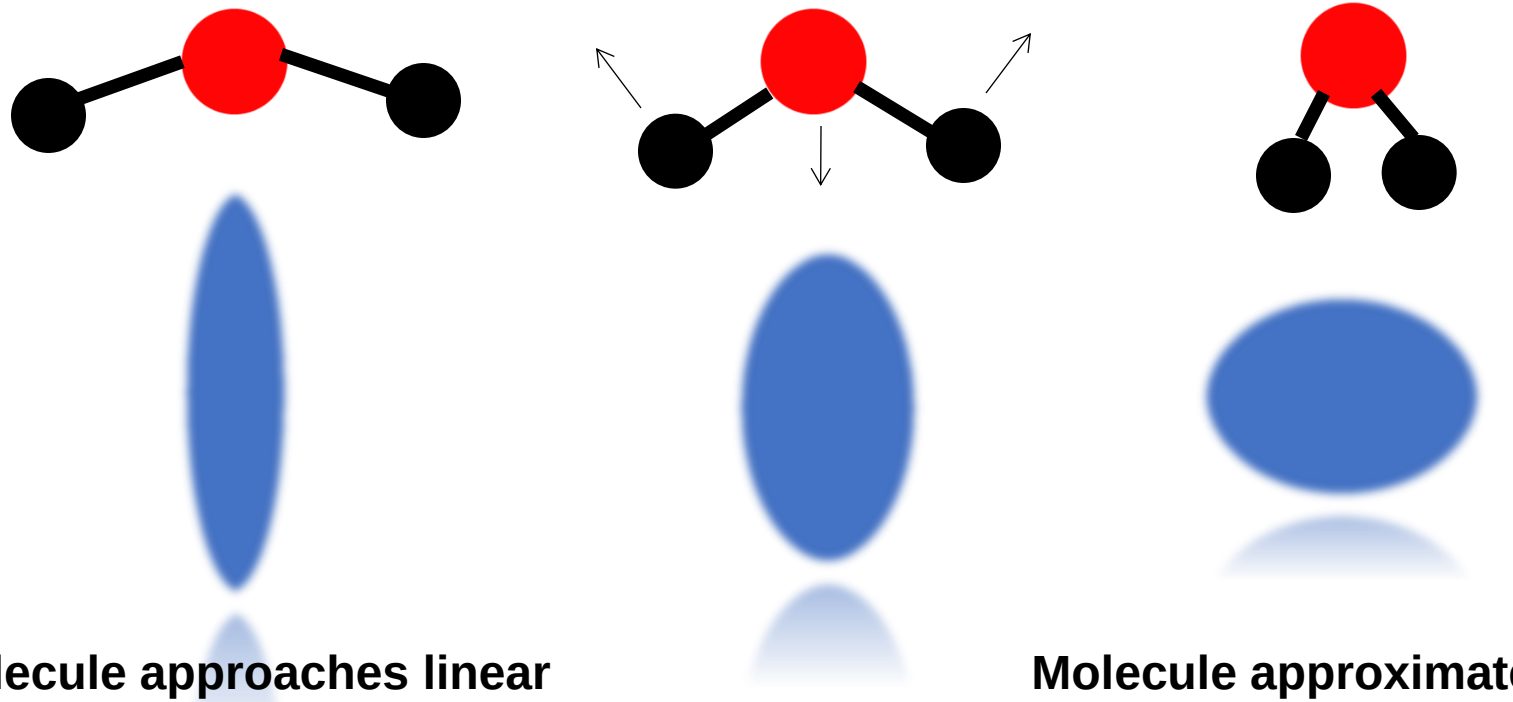
Size of ellipsoid decreases.

Compression → Electrons are more firmly held to the nuclei → Bond becomes less polarizable.

Size of ellipsoid increases.

Polarizability ellipsoid → a three-dimensional surface whose distance from the electrical centre of the molecule is proportional to $1/\sqrt{\alpha_i}$, where α_i is the polarizability along the line joining a point i on the surface of the ellipsoid with the electrical centre of the molecule.

Bending mode in water

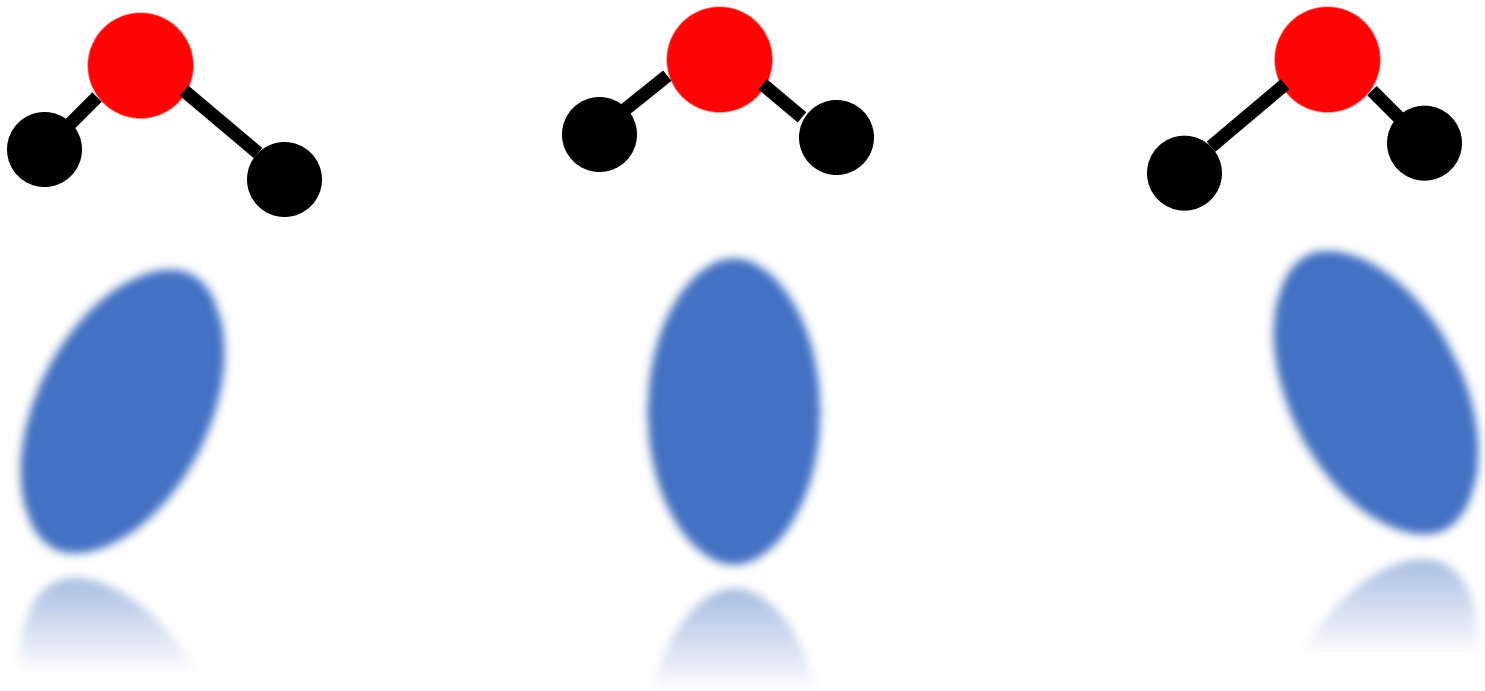


Molecule approaches linear configuration

Molecule approximates to a diatomic molecule

The shape of the ellipsoid changes most in case of bending motions.

Asymmetric stretching mode in water



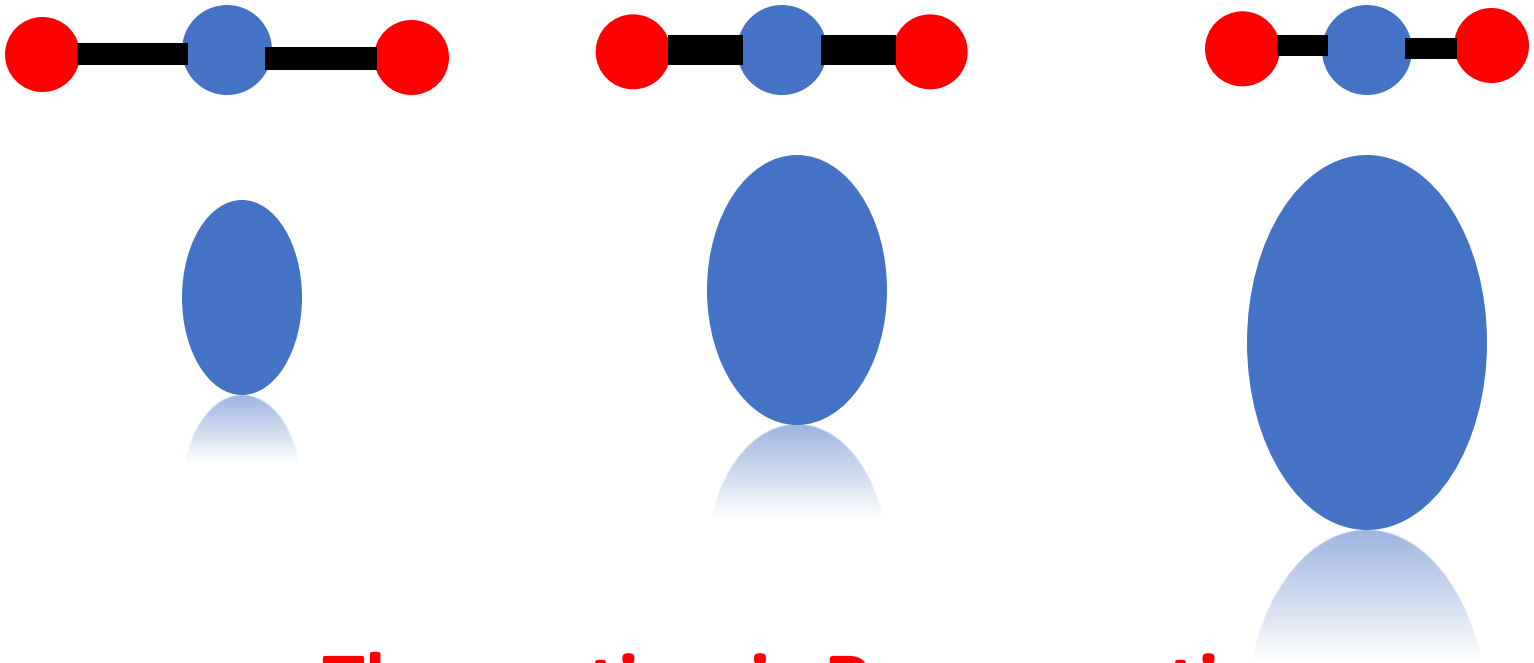
The size and the shape of the ellipsoid remain approximately constant but the direction of the major axis changes.

All the modes are Raman active.

Vibrational Raman Spectra

Carbon dioxide

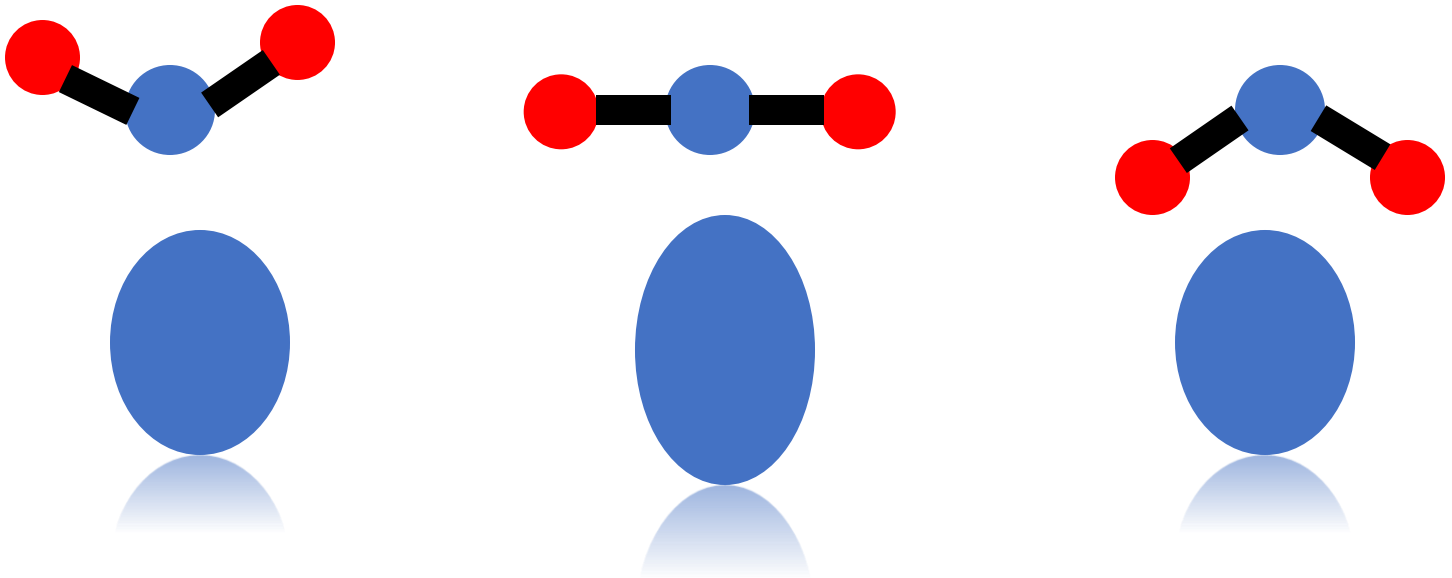
Symmetric stretching mode



The motion is Raman active

Carbon dioxide

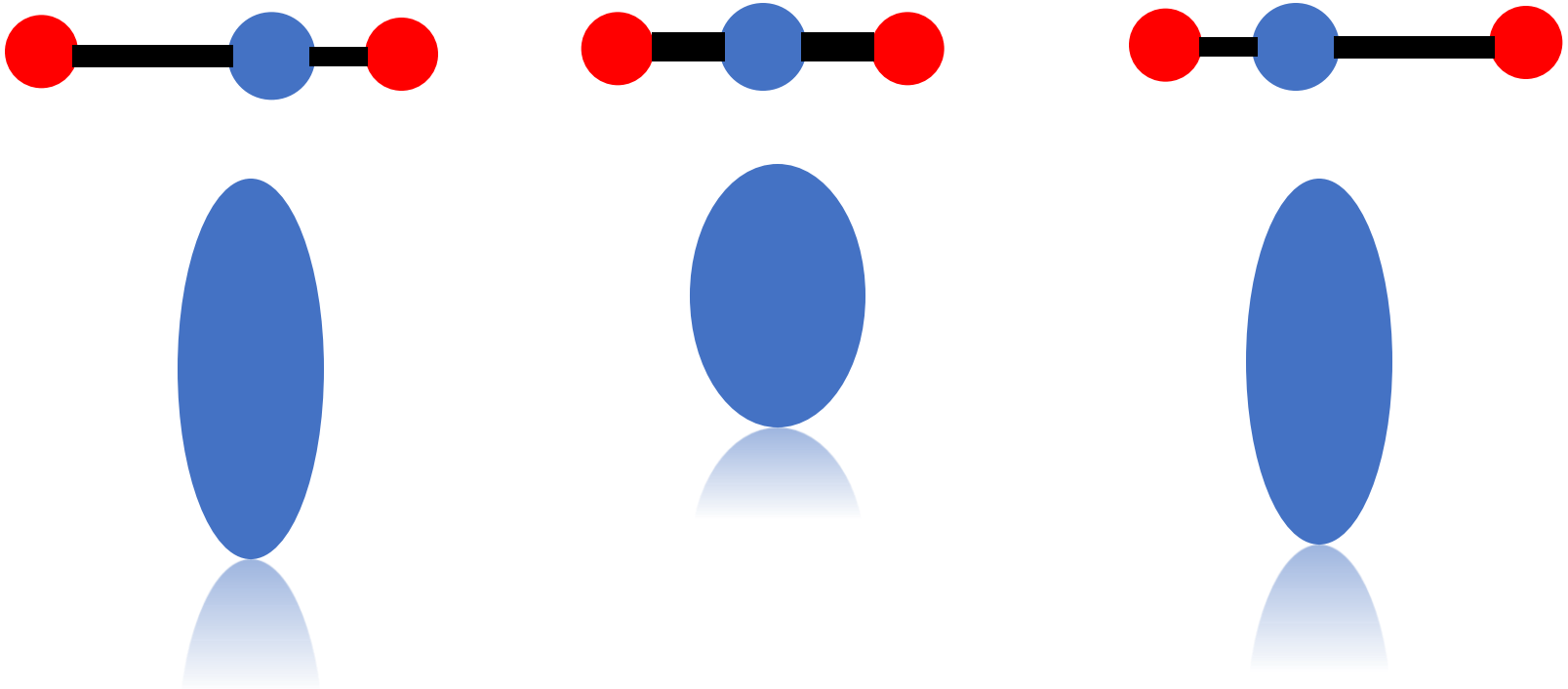
Bending mode



The polarizability ellipsoid changes shape during each vibration. The motion appears to be Raman active.

Carbon dioxide

Asymmetric stretching mode



The polarizability ellipsoid changes shape during each vibration. The motion appears to be Raman active.

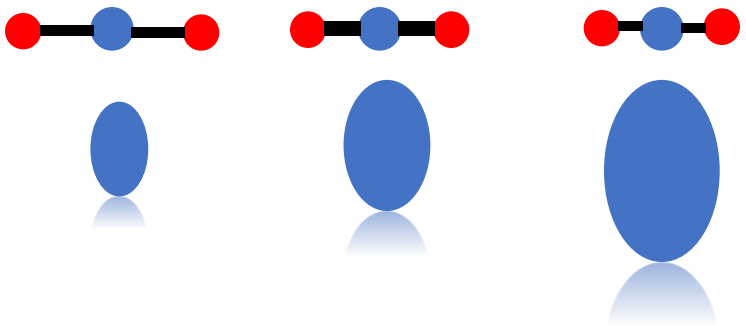
The bending and asymmetric stretching are Raman inactive

Explanation on the basis of change of polarizability with some displacement coordinate (ξ)

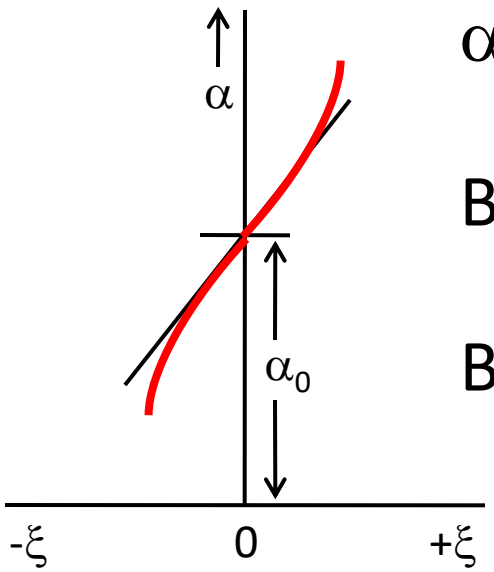
For stretching motion, ξ is a measure of the extension (+ve) or compression (-ve) of the bond.

For bending motion, ξ is measures the displacement (+ve or –ve) of the bond angle from its equilibrium value.

Symmetric stretching mode



Raman active



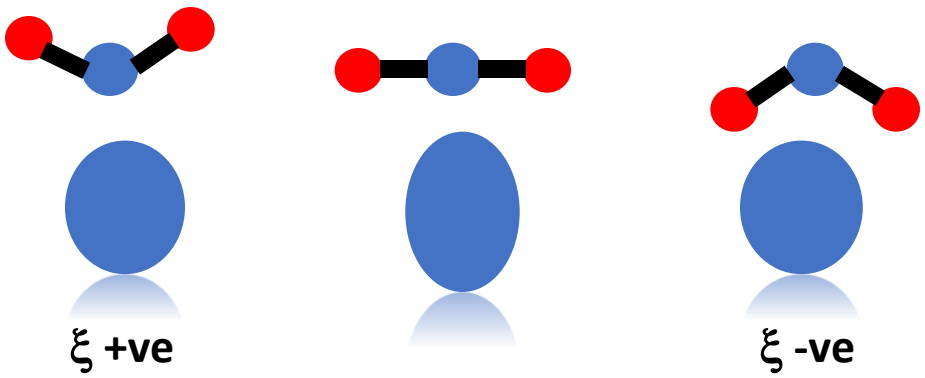
α_0 is the equilibrium polarizability.

Bonds stretch $\rightarrow$ ξ +ve $\rightarrow$ α increases.

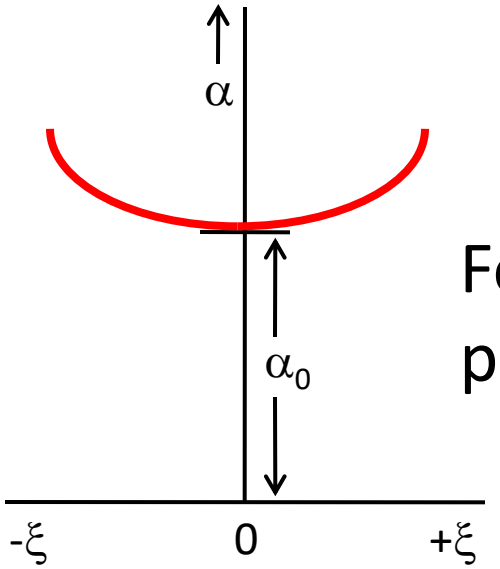
Bonds contract $\rightarrow$ ξ -ve $\rightarrow$ α decreases.

$d\alpha/d\xi \neq 0$ at $\xi = 0$ $\Rightarrow$ For small displacements the motion produces a change in polarizability.

Bending mode



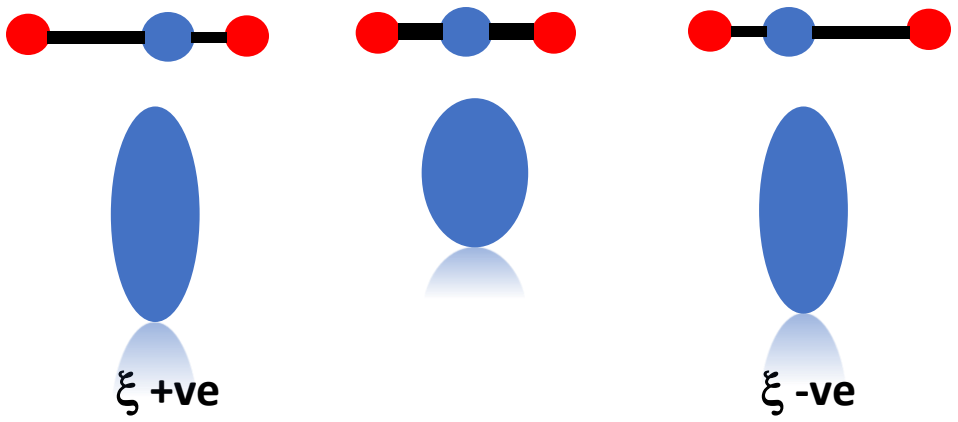
Raman inactive



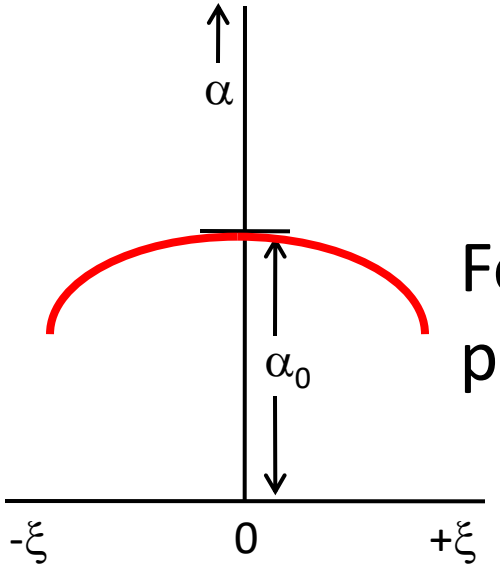
For ξ +ve or ξ -ve $\rightarrow$ The change in polarizability is exactly the same.

$d\alpha/d\xi = 0$ at $\xi = 0$ $\Rightarrow$ For small displacements the motion produces no change in polarizability.

Asymmetric stretching mode



Raman inactive



For ξ +ve or ξ -ve $\rightarrow$ The change in polarizability is exactly the same.

$d\alpha/d\xi = 0$ at $\xi = 0$ $\Rightarrow$ For small displacements the motion produces no change in polarizability.

Rule of Mutual Exclusion

If a molecule has a *centre of symmetry* then Raman active vibrations are infra-red inactive, and vice versa.

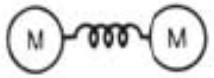
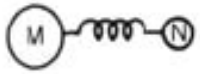
If there is no centre of symmetry then some vibrations may be both Raman and infra-red active.

Raman and infra-red activities for carbon dioxide

Mode of vibration	Raman	Infra-red
Symmetric stretch	Active	Inactive
Bend	Inactive	Active
Asymmetric stretch	Inactive	Active

Raman and infrared active modes

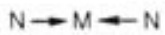
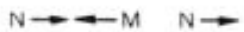
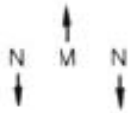
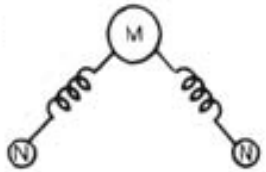
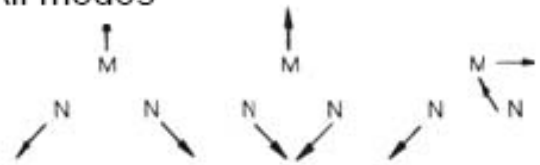
Diatomics

		IR	Raman
Homonuclear		No	Yes
Heteronuclear		Yes	Yes

- A vibrational mode is active for IR-absorption if the **dipole moment** of the molecule **changes** when the atoms are displaced relative to one another (note that the molecule need not have a permanent dipole moment to be IR active).

- A vibrational mode is **Raman-active** if the **polarizability** of the molecule **changes** when the atoms are displaced relative to one another.

Triatomics

		IR	Raman
Symmetric linear			
Symmetric stretch		No	Yes
Asymmetric stretch		Yes	No
Bending		Yes	No
Symmetric non-linear			
All modes		Yes	Yes

The structure of vibrational Raman spectra

For every vibrational mode, we have:

$$\varepsilon = \overline{\omega}_e\left(\upsilon + \frac{1}{2}\right) - \overline{\omega}_e x_e\left(\upsilon + \frac{1}{2}\right)^2 \quad \mathrm{cm}^{-1} \quad (\upsilon = 0,1,2,...)$$

The selection rule is: $\Delta\upsilon = 0, \pm 1, \pm 2, ...$

Same for Raman and IR.

For Raman active modes:

$$\upsilon = 0 \rightarrow \upsilon = 1: \quad \Delta\varepsilon_{\text{fundamental}} = \overline{\omega}_e \left(1 - 2x_e\right) \quad \text{cm}^{-1}$$

$$\upsilon = 0 \rightarrow \upsilon = 2: \quad \Delta\varepsilon_{\text{overtone}} = 2\overline{\omega}_e \left(1 - 3x_e\right) \quad \text{cm}^{-1}$$

$$\upsilon = 1 \rightarrow \upsilon = 2: \quad \Delta\varepsilon_{\text{hot}} = \overline{\omega}_e \left(1 - 4x_e\right) \quad \text{cm}^{-1}$$

These lines are very weak in intensity and in Raman spectroscopy they may not be observed.

Rotational Fine Structure

Diatomic Molecules

The vibration-rotation energy levels are given by:

$$\varepsilon_{J,\upsilon} = \overline{\omega}_e\left(\upsilon + \frac{1}{2}\right) - \overline{\omega}_e x_e\left(\upsilon + \frac{1}{2}\right)^2 + BJ\left(J+1\right) \quad \mathrm{cm}^{-1}$$

$$(\upsilon = 0,1,2,\ldots;\ \mathrm{J} = 0,1,2,\ldots)$$

We ignore the centrifugal distortion.

Selection rule: ΔJ = 0, $\pm$2

Selection rule: $\Delta J = 0, \pm 2$

$$\bar{\nu}_0 = \bar{\omega}_e(1 - 2x_e)$$

For $\upsilon = 0 \rightarrow \upsilon = 1$

$$\Delta J = 0: \quad \Delta\varepsilon_{\mathrm{Q}} = \bar{\nu}_0 \ \ \mathrm{cm}^{-1} \qquad \text{(For all J)}$$

$$\Delta J = +2: \quad \Delta\varepsilon_{\mathrm{S}} = \bar{\nu}_0 + \mathrm{B}(4\mathrm{J} + 6) \ \ \mathrm{cm}^{-1} \qquad (\mathrm{J} = 0,1,2,\ldots)$$

$$\Delta J = -2: \quad \Delta\varepsilon_{\mathrm{O}} = \bar{\nu}_0 - \mathrm{B}(4\mathrm{J} + 6) \ \ \mathrm{cm}^{-1} \qquad (\mathrm{J} = 2,3,4,\ldots)$$

Stokes lines (low frequency) will appear at:

$$\bar{\nu}_Q = \bar{\nu}_{ex} - \Delta\varepsilon_{\mathrm{Q}} = \bar{\nu}_{ex} - \bar{\nu}_0 \ \ \mathrm{cm}^{-1} \qquad \text{(For all J)}$$

$$\bar{\nu}_S = \bar{\nu}_{ex} - \Delta\varepsilon_{\mathrm{S}} = \bar{\nu}_{ex} - \bar{\nu}_0 - \mathrm{B}(4\mathrm{J} + 6) \ \ \mathrm{cm}^{-1} \qquad (\mathrm{J} = 0,1,2,\ldots)$$

$$\bar{\nu}_O = \bar{\nu}_{ex} - \Delta\varepsilon_{\mathrm{O}} = \bar{\nu}_{ex} - \bar{\nu}_0 + \mathrm{B}(4\mathrm{J} + 6) \ \ \mathrm{cm}^{-1} \qquad (\mathrm{J} = 2,3,4,\ldots)$$

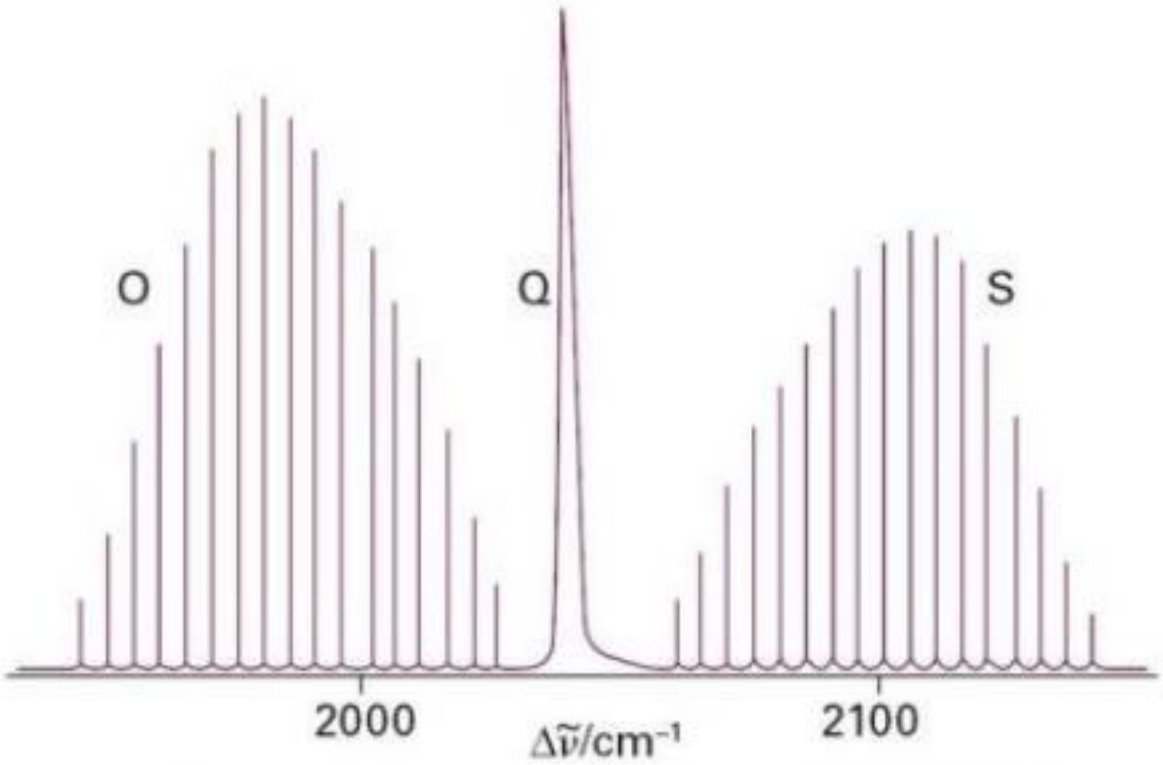
Theoretically it is possible for vibrational and rotational transitions to occur simultaneously in a Raman transition.

- In such a case, the selection rules are identical with those for separate rotational and vibrational transitions, i.e., $\Delta J = +2, 0$ or -2 and $\Delta v = +1$ or -1.
- Since ΔJ may be zero, a Raman line representing Q branch will be observed.
- In a nomenclature, spectral lines are designated O, P, Q, R and S branches corresponding to ΔJ values of -2, -1, 0, + 1 and +2, respectively.
- The frequency of this line is referred to as $\Delta\nu_0$ and identical with that for the pure vibrational transition.
- If $\Delta\nu_0$ is the frequency shift for the purely vibrational transition (Q branch), the Raman shifts for the accompanying rotational transitions will be represented by

  $\Delta J = +2$; $\Delta\nu = \Delta\nu_0 + 2B(2J+3)$, where J=0,1,2,..... and

  $\Delta J = -2$; $\Delta\nu = \Delta\nu_0 - 2B(2J+3)$, where J=0,1,2,.....
- Thus, the central vibrational Raman lines (Q branch) will be accompanied by two wings or branches. The branch on the low frequency side for which ΔJ is -2 is known as O branch, and that on the high frequency side for which ΔJ is +2 is called S branch.

Vibration-rotation Raman spectrum of CO



ΔJ=-2 ΔJ=0 ΔJ=+2

•**Selection rules:**

Unlike infrared spectroscopy, which relies on dipole moment changes, Raman spectroscopy observes transitions where the molecule's polarizability changes during vibration or rotation, meaning certain molecules that are IR inactive can be studied using Raman.

•**Spectrum features:**

The Raman spectrum exhibits a series of peaks corresponding to different vibrational modes, and within each vibrational peak, fine structure is observed due to rotational transitions, allowing for detailed analysis of both vibrational and rotational energy levels.

•**Applications:**

•Identifying functional groups in molecules

•Studying molecular symmetry

•Determining bond lengths and angles

•Analyzing chemical reactions

•Studying biological samples due to minimal water interference

RULE OF MUTUAL EXCLUSION

This rule states that

if a molecule has a center of symmetry,
then Raman active vibrations are IR inactive and IR active vibrations are Raman inactive.

If there is no center of symmetry,
then some (but not necessarily all) vibrations may be both Raman and IR active.

Thus, the Raman and IR spectra having no common bands show that the molecule **has a center of symmetry** but here caution is necessary because a vibration may be Raman active but too weak to be observed.

If some of the bands are present in both the Raman and IR spectra, it is certain that the molecule has **no center of symmetry**.

RAMAN SPECTRA	IR SPECTRA
1. These originate from scattering of radiation by vibrating and rotating molecules.	1. These originate from absorption of radiation by vibrating and rotating molecules.
2. Some change in molecular polarizability during the molecular vibration or rotation is essential for a molecule to exhibit Raman spectrum.	2. Some change in dipole moment during the molecular vibration or rotation is essential for a molecule to exhibit IR spectrum.
3. Raman lines are weak in intensity, hence concentrated solutions are preferred as samples to give enough intensity.	3. Generally, dilute solutions are preferred.
4. Water used as solvent.	4. Water cannot be used
5. Optical systems of Raman spectrometer made of glass or quartz.	5. Optical systems of IR spectrometer made of NaCl, NaBr, KCl, KBr, etc.
6. Sometimes photochemical reactions take place in the frequency regions of Raman lines and so create difficulties.	6. Photochemical reactions do not take place.
7. These are recorded by using a beam of monochromatic radiation.	7. These are recorded by using a beam of radiation having a large number of frequencies in the IR region.
8. Homonuclear diatomic molecules are Raman active.	8. Homonuclear diatomic molecules are IR inactive.
9. Pure substances are required for studies by Raman spectra.	9. Studies by the IR spectra do not require a high degree of purity.